Name:	Student ID:

GAUSSIAN INTEGRALS AND GAUSSIAN PROCESSES PROBLEM SHEET

This sheet goes with the lectures on Gaussian integrals and Gaussian processes. Questions 1 and 2 only ask you to *use* the results below; Questions 3 and 4 derive them, and are longer and harder. Do Questions 1 and 2 first.

The Gaussian toolkit

- **Integrals:** $\displaystyle\int_{-\infty}^{\infty} \mathrm{e}^{-\frac{a}{2}x^2+bx}\,\mathrm{d}x = \sqrt{\frac{2\pi}{a}}\,\mathrm{e}^{\frac{b^2}{2a}}$ for $a > 0$, and in n dimensions, for positive definite $\mathbf{A}$, $\displaystyle\int \mathrm{e}^{-\frac{1}{2}\boldsymbol{x}^\mathsf{T}\mathbf{A}\boldsymbol{x}+\boldsymbol{b}^\mathsf{T}\boldsymbol{x}}\,\mathrm{d}\boldsymbol{x} = \sqrt{\frac{(2\pi)^n}{\det(\mathbf{A})}}\,\mathrm{e}^{\frac{1}{2}\boldsymbol{b}^\mathsf{T}\mathbf{A}^{-1}\boldsymbol{b}}$.
- **Moment generating function:** if $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu},\boldsymbol{\Sigma})$ then $\mathbb{E}\left[\mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{x}}\right] = \mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{\mu}+\frac{1}{2}\boldsymbol{t}^\mathsf{T}\boldsymbol{\Sigma}\boldsymbol{t}}$ (in one dimension $\mathbb{E}\left[\mathrm{e}^{tX}\right] = \mathrm{e}^{\mu t+\frac{1}{2}\sigma^2 t^2}$).
- **Sums:** if $X \sim \mathcal{N}(\mu_1,\sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2,\sigma_2^2)$ are independent then $X + Y \sim \mathcal{N}(\mu_1+\mu_2,\sigma_1^2+\sigma_2^2)$.
- **Linear maps:** $\mathbf{W}\boldsymbol{x}+\boldsymbol{c} \sim \mathcal{N}(\mathbf{W}\boldsymbol{\mu}+\boldsymbol{c},\ \mathbf{W}\boldsymbol{\Sigma}\mathbf{W}^\mathsf{T})$.
- **Marginals and conditionals:** for $\begin{pmatrix}\boldsymbol{z}_1\\ \boldsymbol{z}_2\end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{pmatrix}, \begin{pmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12}\\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22}\end{pmatrix}\right)$:
  - marginal: $\boldsymbol{z}_1 \sim \mathcal{N}(\boldsymbol{\mu}_1,\boldsymbol{\Sigma}_{11})$ (strike out the rows and columns of $\boldsymbol{z}_2$);
  - conditional: $\boldsymbol{z}_1 \mid \boldsymbol{z}_2 \sim \mathcal{N}\big(\boldsymbol{\mu}_1+\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\boldsymbol{z}_2-\boldsymbol{\mu}_2),\ \boldsymbol{\Sigma}_{11}-\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\big)$.

1 Each part needs only one or two of the results in the box.

(a) Evaluate $\displaystyle\int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2+3x}\,\mathrm{d}x$. [2 marks]

$\overline{2}$

(b) $X \sim \mathcal{N}(1,4)$. Use the moment generating function to find $\mathbb{E}\left[\mathrm{e}^X\right]$, and $\mathbb{E}\left[X^2\right]$. [3 marks]

3

(c) $X \sim \mathcal{N}(1,1)$ and $Y \sim \mathcal{N}(2,3)$ are independent. What are the distributions of $X+Y$ and of $2X-Y$? [3 marks]

3

(d) $\boldsymbol{x} \sim \mathcal{N}(\mathbf{0},\mathbf{I})$ in two dimensions and $\boldsymbol{z} = \mathbf{W}\boldsymbol{x}$ with $\mathbf{W} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. What is the distribution of $\boldsymbol{z}$? Are $z_1 = x_1 + x_2$ and $z_2 = x_1 - x_2$ independent? [3 marks]

3

(e) $\boldsymbol{z} \sim \mathcal{N}(\boldsymbol{\mu},\boldsymbol{\Sigma})$ with $\boldsymbol{\mu} = (1,2,3)^\mathsf{T}$ and $\boldsymbol{\Sigma} = \begin{pmatrix} 4 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}$. What is the marginal distribution of (z_1, z_3)? [2 marks]

2

(f) (z_1, z_2) is Gaussian with mean $\mathbf{0}$ and covariance $\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$. We observe $z_2 = 2$. What is the distribution of z_1 now? Compare it with the distribution of z_1 before the observation. [3 marks]

$\frac{}{3}$

(g) Our prior belief about a quantity is $x \sim \mathcal{N}(0, 4)$. We measure $y = x + \epsilon$ with noise $\epsilon \sim \mathcal{N}(0, 1)$ and get $y = 3$. Write down the joint distribution of (x, y) and hence find the distribution of x given $y = 3$. [4 marks]

$\frac{}{4}$

End of question 1

(a) $\frac{}{2}$	(b) $\frac{}{3}$	(c) $\frac{}{3}$	(d) $\frac{}{3}$	(e) $\frac{}{2}$	(f) $\frac{}{3}$	(g) $\frac{}{4}$	Total $\frac{}{20}$

2 A Gaussian process with covariance function $K(x,x') = \mathrm{e}^{-(x-x')^2/2}$ has, for observations $\boldsymbol{y}$ at inputs $x_1,\ldots,x_m$ with noise variance σ^2, the predictive distribution at a new input x_*

$$f_* \mid \boldsymbol{y} \sim \mathcal{N}\big(\boldsymbol{k}_*^\mathsf{T}(\mathbf{K}+\sigma^2\mathbf{I})^{-1}\boldsymbol{y},\; k_* - \boldsymbol{k}_*^\mathsf{T}(\mathbf{K}+\sigma^2\mathbf{I})^{-1}\boldsymbol{k}_*\big)$$

where $K_{ij} = K(x_i,x_j)$, $(\boldsymbol{k}_*)_i = K(x_i,x_*)$ and $k_* = K(x_*,x_*)$.

(a) Explain why this is the conditional result in the box, with $\boldsymbol{z}_1 = f_*$ and $\boldsymbol{z}_2 = \boldsymbol{y}$. What are $\boldsymbol{\Sigma}_{11}$, $\boldsymbol{\Sigma}_{12}$ and $\boldsymbol{\Sigma}_{22}$? [3 marks]

3

(b) Take a single noise-free observation ($\sigma^2 = 0$), $y_1 = 1$ at $x_1 = 0$. Find the predictive mean and variance at x_*. What are they at $x_* = 0$, and as $x_* \to \infty$? [4 marks]

4

(c) Repeat with noise $\sigma^2 = 1$. What changes at $x_* = 0$? [3 marks]

3

(d) Now take two noise-free observations, $y = 1$ at $x = -1$ and $y = 1$ at $x = 1$. Find the predictive mean and variance at $x_* = 0$. (Hint: $\boldsymbol{y} = (1,1)^\mathsf{T}$ is an eigenvector of **K**.) [5 marks]

5

End of question 2

(a) $\frac{}{3}$ (b) $\frac{}{4}$ (c) $\frac{}{3}$ (d) $\frac{}{5}$ Total $\frac{}{15}$

3 This question derives the normalisation of the multivariate normal step by step.

(a) Consider the integral

$$I_1 = \int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2} \mathrm{d}x.$$

Directly evaluating this is difficult, but there is a trick. Consider instead

$$I_1^2 = \int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2} \mathrm{d}x \int_{-\infty}^{\infty} \mathrm{e}^{-y^2/2} \mathrm{d}y = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathrm{e}^{-(x^2+y^2)/2} \mathrm{d}x \mathrm{d}y.$$

By making the change of variables to polar coordinates where $r = \sqrt{x^2 + y^2}$ and $\theta = \arctan(y/x)$ (so that $x = r\cos(\theta)$, $y = r\sin(\theta)$) then $\mathrm{d}x\mathrm{d}y = r\mathrm{d}r\mathrm{d}\theta$. Note that to integrate over all space we let θ vary from 0 to 2π and r to vary from 0 to ∞. Write down the integral in polar coordinates, make the further change of variables $u = r^2/2$ to evaluate I_1^2, and hence compute I_1.

[5 marks]

5

(b) By making a change of variables compute

$$I_2 = \int_{-\infty}^{\infty} \mathrm{e}^{-(x-\mu)^2/(2\sigma^2)} \mathrm{d}x$$

[5 marks]

5

(c) By using the identity $\mathrm{e}^{a+b} = \mathrm{e}^a \mathrm{e}^b$, or more generally

$$\mathrm{e}^{\sum_i a_i} = \prod_i \mathrm{e}^{a_i}$$

compute

$$I_3 = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{1}{2}\|\boldsymbol{x}\|_2^2} \mathrm{d}x_1 \cdots \mathrm{d}x_n$$

where $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)^{\mathsf{T}}$. [3 marks]

3

(d) By using the fact that for a positive definite matrix, $\boldsymbol{\Sigma}$, we can use the eigenvector decomposition $\boldsymbol{\Sigma}^{-1} = \mathbf{V}\boldsymbol{\Lambda}^{-1}\mathbf{V}^{\mathsf{T}}$ where $\mathbf{V}$ is an orthogonal matrix with determinant $\det(\mathbf{V}) = \pm 1$ and $\boldsymbol{\Lambda}^{-1}$ is a diagonal matrix with elements λ_i^{-1} compute

$$I_4 = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})} \mathrm{d}x_1 \cdots \mathrm{d}x_n.$$

[6 marks]

6

• Do not write in this space •

(e) Using the facts, that $\mathbf{\Sigma} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{\mathsf{T}}$, for any two square matrices **A** and **B** the determinants satisfy $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$, and $\det(\mathbf{V}) = \det(\mathbf{V}^{\mathsf{T}}) = \pm 1$ show that $\det(\mathbf{\Sigma}) = \prod_i \lambda_i$. [1 mark]

$\frac{}{1}$

End of question 3

(a) $\frac{}{5}$	(b) $\frac{}{5}$	(c) $\frac{}{3}$	(d) $\frac{}{6}$	(e) $\frac{}{1}$	Total $\frac{}{20}$

4 This question is harder: it derives the marginal and conditional results in the box by doing the integrals directly (the lecture used a shorter route). Take it slowly; each part is a short calculation.

Consider a multivariate normal distribution

$$f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y}) = \mathcal{N}\left(\begin{pmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{pmatrix} \middle| \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^{\mathsf{T}} & \mathbf{C} \end{pmatrix} \right)$$

where **A** and **C** are symmetric (positive definite) matrices. The matrix

$$\boldsymbol{\Sigma} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^{\mathsf{T}} & \mathbf{C} \end{pmatrix}$$

is the covariance matrix.

We want to compute the conditional probability density function $f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x} \mid \boldsymbol{y})$. This is complicated because the normal distribution involves the inverse of the covariance matrix. Let

$$\mathbf{U} = \begin{pmatrix} \mathbf{I} & \mathbf{B}\mathbf{C}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}, \qquad \mathbf{D} = \begin{pmatrix} \mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}} & \mathbf{0} \\ \mathbf{0}^{\mathsf{T}} & \mathbf{C} \end{pmatrix}$$

where $\mathbf{I}$ is the identity matrix.

(a) Compute $\mathbf{UD}$ [3 marks]

3

(b) Using the previous result compute $(\mathbf{UD})\mathbf{U}^{\mathsf{T}}$. Hence show $\boldsymbol{\Sigma} = \mathbf{UDU}^{\mathsf{T}}$. [3 marks]

3

(c) Given that $\mathbf{\Sigma} = \mathbf{U}\mathbf{D}\mathbf{U}^{\mathsf{T}}$ write down $\mathbf{\Sigma}^{-1}$ in terms of $\mathbf{U}$ and $\mathbf{D}$ [1 mark]

$\overline{1}$

(d) Demonstrate by direct multiplication that

$$\mathbf{U}^{-1} = \begin{pmatrix} \mathbf{I} & -\mathbf{B}\mathbf{C}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \quad \mathbf{D}^{-1} = \begin{pmatrix} (\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})^{-1} & \mathbf{0} \\ \mathbf{0}^{\mathsf{T}} & \mathbf{C}^{-1} \end{pmatrix} \quad (\mathbf{U}^{\mathsf{T}})^{-1} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}} & \mathbf{I} \end{pmatrix}$$

i.e. show $\mathbf{U}^{-1}\mathbf{U} = \mathbf{I}$, $\mathbf{D}^{-1}\mathbf{D} = \mathbf{I}$ and $(\mathbf{U}^{\mathsf{T}})^{-1}\mathbf{U}^{\mathsf{T}} = \mathbf{I}$.

[6 marks]

i

ii

iii

$\overline{6}$

(e) Letting $\boldsymbol{z} = \binom{\boldsymbol{x}}{\boldsymbol{y}}$ then we can write

$$f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y}) = f_{\boldsymbol{Z}}(\boldsymbol{z}) = \mathcal{N}(\boldsymbol{z} \mid \boldsymbol{0},\boldsymbol{\Sigma}) = \frac{1}{\sqrt{\det(2\pi\boldsymbol{\Sigma})}} \mathrm{e}^{-\frac{1}{2}\boldsymbol{z}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\boldsymbol{z}}.$$

where $\det(2\pi\boldsymbol{\Sigma})$ is the determinant of the matrix $2\pi\boldsymbol{\Sigma}$ and is introduced to ensure that $f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y})$ is normalised. From parts (c) and (d)

$$\boldsymbol{\Sigma}^{-1} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}} & \mathbf{I} \end{pmatrix} \begin{pmatrix} (\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}})^{-1} & \mathbf{0} \\ \mathbf{0}^{\mathsf{T}} & \mathbf{C}^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{I} & -\mathbf{B}\mathbf{C}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}.$$

Expand out $\boldsymbol{z}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\boldsymbol{z} = (\boldsymbol{x}^{\mathsf{T}},\boldsymbol{y}^{\mathsf{T}})\boldsymbol{\Sigma}^{-1}\binom{\boldsymbol{x}}{\boldsymbol{y}}$ (start by multiplying the vectors $\boldsymbol{z}^{\mathsf{T}}$ by $(\mathbf{U}^{\mathsf{T}})^{-1}$ and $\boldsymbol{z}$ by $\mathbf{U}^{-1}$)

[4 marks]

$\boxed{4}$

(f) (We write $\int f(\boldsymbol{z})\mathrm{d}\boldsymbol{z}$ for $\int\cdots\int f(\boldsymbol{z})\mathrm{d}z_1\cdots\mathrm{d}z_n$.) To compute $f_{\boldsymbol{X}|\boldsymbol{Y}}(\boldsymbol{x} \mid \boldsymbol{y}) = f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y})/f_{\boldsymbol{Y}}(\boldsymbol{y})$ we need to find

$$\begin{aligned} f_{\boldsymbol{Y}}(\boldsymbol{y}) &= \int_{-\infty}^{\infty} f(\boldsymbol{x},\boldsymbol{y})\mathrm{d}\boldsymbol{x} \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{\det(2\pi\boldsymbol{\Sigma})}} \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}^\mathsf{T}-\boldsymbol{y}^\mathsf{T}\mathbf{C}^{-1}\mathbf{B}^\mathsf{T})(\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^\mathsf{T})^{-1}(\boldsymbol{x}-\mathbf{B}\mathbf{C}^{-1}\boldsymbol{y})-\frac{1}{2}\boldsymbol{y}^\mathsf{T}\mathbf{C}^{-1}\boldsymbol{y}}\mathrm{d}\boldsymbol{x} \\ &= \frac{\mathrm{e}^{-\frac{1}{2}\boldsymbol{y}^\mathsf{T}\mathbf{C}^{-1}\boldsymbol{y}}}{\sqrt{\det(2\pi\boldsymbol{\Sigma})}} \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}^\mathsf{T}-\boldsymbol{y}^\mathsf{T}\mathbf{C}^{-1}\mathbf{B}^\mathsf{T})(\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^\mathsf{T})^{-1}(\boldsymbol{x}-\mathbf{B}\mathbf{C}^{-1}\boldsymbol{y})}\mathrm{d}\boldsymbol{x} \end{aligned}$$

By making a change of variable from $\boldsymbol{x}$ to $\boldsymbol{u} = \boldsymbol{x} - \mathbf{B}\mathbf{C}^{-1}\boldsymbol{y}$ rewrite the integral

$$I = \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}^\mathsf{T}-\boldsymbol{y}^\mathsf{T}\mathbf{C}^{-1}\mathbf{B}^\mathsf{T})(\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^\mathsf{T})^{-1}(\boldsymbol{x}-\mathbf{B}\mathbf{C}^{-1}\boldsymbol{y})}\mathrm{d}\boldsymbol{x}$$

then use

$$\int_{-\infty}^{\infty} \mathrm{e}^{-\frac{1}{2}\boldsymbol{u}^\mathsf{T}\mathbf{M}^{-1}\boldsymbol{u}}\mathrm{d}\boldsymbol{u} = \sqrt{\det(2\pi\mathbf{M})}$$

to evaluate $f_{\boldsymbol{Y}}(\boldsymbol{y})$.

[5 marks]

5

(g) Using $f_{\boldsymbol{X}|\boldsymbol{Y}}(\boldsymbol{x} \mid \boldsymbol{y}) = f_{\boldsymbol{X},\boldsymbol{Y}}(\boldsymbol{x},\boldsymbol{y})/f_{\boldsymbol{Y}}(\boldsymbol{y})$ write down $f_{\boldsymbol{X}|\boldsymbol{Y}}(\boldsymbol{x} \mid \boldsymbol{y})$. [3 marks]

3

End of question 4

(a)	(b)	(c)	(d)	(e)	(f)	(g)	Total
$\frac{}{3}$	$\frac{}{3}$	$\frac{}{1}$	$\frac{}{6}$	$\frac{}{4}$	$\frac{}{5}$	$\frac{}{3}$	$\frac{}{25}$